Lab Report

Module 2 — Network Discovery and Scanning with Nmap

Cybersecurity Hands-On Lab

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1. Experiment Description

This lab was about network reconnaissance — the first step attackers take before launching any real attack. We used Nmap (Network Mapper) to scan a local network, find active devices, check open ports, identify running services, and detect the operating system of a target host.

Everything was done on the campus Wi-Fi network (192.168.11.0/24) using a Windows 11 PC with Nmap 7.98 installed. The target machine for the detailed scans was 192.168.11.1, which turned out to be the network router (ZTE brand, running Linux).

2. Commands Used

Command	Purpose
`nmap --version`	Verify Nmap is installed and check version
`nmap -sn 192.168.11.0/24`	Ping scan to discover active hosts on the network
`nmap 192.168.11.1`	Basic port scan on the target host
`nmap -sV 192.168.11.1`	Service and version detection
`nmap -O 192.168.11.1`	Operating system detection
`nmap -A 192.168.11.1`	Aggressive scan (OS + services + scripts + traceroute)

3. Screenshots and Results

3.1 Nmap version check

First thing I did was verify that Nmap was properly installed. The command nmap --version confirms we're running Nmap 7.98, compiled for Windows (i686-pc-windows-windows) with OpenSSL 3.0.17 and Npcap 1.86 support.

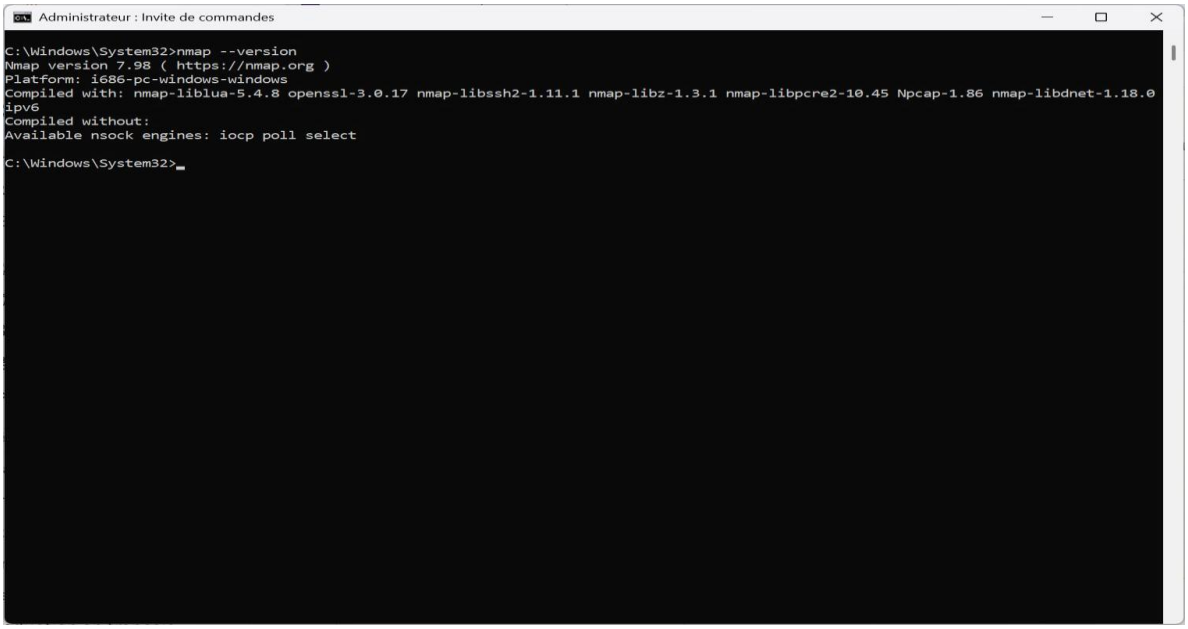


Figure 1 — nmap --version: Nmap 7.98 confirmed on Windows

3.2 Host discovery — nmap -sn 192.168.11.0/24

The ping scan found 6 active hosts on the network in just 3.83 seconds. The first attempt on 192.168.1.0/24 had returned 0 results because that wasn't our actual network range — running ipconfig first revealed the correct subnet was 192.168.11.0/24.

Active hosts discovered:

- 192.168.11.1 — MAC: 68:9E:29:3A:F5:20 (ZTE) — this is the router/gateway
- 192.168.11.100 — MAC: 62:3E:5D:22:5A:0F (Unknown)
- 192.168.11.101 — MAC: E6:1F:56:46:07:28 (Unknown)
- 192.168.11.102 — MAC: D2:C9:57:84:F9:AC (Unknown)
- 192.168.11.103 — this machine (no MAC shown, it's our own PC)
- 192.168.11.104 — MAC: CA:AE:EA:B4:CD:D4 (Unknown)

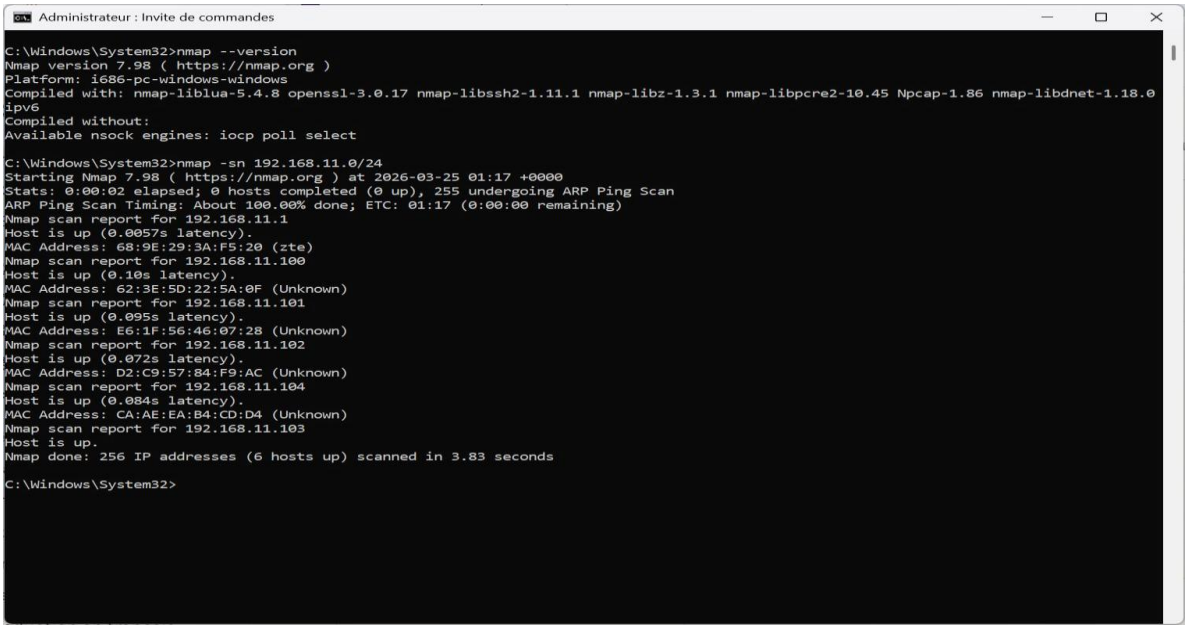


Figure 2 — nmap -sn 192.168.11.0/24: 6 active hosts discovered

3.3 Port scan — nmap 192.168.11.1

Scanning the router directly revealed 5 ports of interest out of 1000 scanned. 995 were closed. Here's what came back:

Port	State	Service	Notes
`23/tcp`	filtered	telnet	Blocked — old remote access protocol
`53/tcp`	open	domain	DNS service running
`80/tcp`	open	http	Web interface accessible
`443/tcp`	open	https	Secure web interface
`52869/tcp`	open	unknown	UPnP service (ZTE-specific)

```
:\Windows\System32>nmap 192.168.11.1
tarting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 01:17 +0000
map scan report for 192.168.11.1
ost is up (0.0054s latency).
ot shown: 995 closed tcp ports (reset)
ORT      STATE    SERVICE
3/tcp    filtered telnet
3/tcp    open     domain
0/tcp    open     http
43/tcp   open     https
2869/tcp open     unknown
AC Address: 68:9E:29:3A:F5:20 (zte)

map done: 1 IP address (1 host up) scanned in 2.54 seconds

:\Windows\System32>
```

Figure 3 — nmap 192.168.11.1: open ports on the router

3.4 Service detection — nmap -sV 192.168.11.1

The -sV flag tells Nmap to probe each open port and try to identify the exact service and version running. The output was very verbose for port 52869, which turned out to be a UPnP service running on a Linux stack. The router's HTTP interface returned a 302 redirect, and the HTTPS interface returned a full login page with security headers like Strict-Transport-Security and X-Frame-Options: SAMEORIGIN.

```
SF:\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20<h2>400\x20Bad\x20Reque
SF:st</h2>\nYour\x20request\x20has\x20bad\x20syntax\x20or\x20is\x20inheren
SF:tly\x20impossible\x20to\x20satisfy\.\n<div\x20style=\"display:none\">\n
SF:<ajax_response_xml_root>\n<IF_ERRORSTR>SessionTimeout</IF_ERRORSTR>\n<I
SF:F_ERRORPARAM>SUCC</IF_ERRORPARAM>\n<IF_ERRORTYPE>SUCC</IF_ERRORTYPE>\n<
SF:/ajax_response_xml_roo");
==============NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)==============
SF-Port52869-TCP:V=7.98%I=7%D=3/25%Time=69C337F9%P=i686-pc-windows-windows
SF:%r(HTTPOptions,BF,"HTTP/0\.0\x20501\x20Not\x20Implemented\r\nSERVER:\x2
SF:0Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20devices/2\.0
SF:\r\nCONTENT-LENGTH:\x2054\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<html><b
SF:ody><h1>501\x20Not\x20Implemented</h1></body></html>")%r(RTSPRequest,B7
SF:,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERVER:\x20Linux,\x20UPnP/1\.0,
SF:\x20Portable\x20SDK\x20for\x20UPnP\x20devices/2\.0\r\nCONTENT-LENGTH:\x
SF:2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<html><body><h1>400\x20Bad\x2
SF:0Request</h1></body></html>")%r(RPCCheck,B7,"HTTP/0\.0\x20400\x20Bad\x2
SF:0Request\r\nSERVER:\x20Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x20for\x
SF:20UPnP\x20devices/2\.0\r\nCONTENT-LENGTH:\x2050\r\nCONTENT-TYPE:\x20tex
SF:t/html\r\n\r\n<html><body><h1>400\x20Bad\x20Request</h1></body></html>"
SF:)%r(DNSVersionBindReqTCP,B7,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERV
SF:ER:\x20Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20device
SF:s/2\.0\r\nCONTENT-LENGTH:\x2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<h
SF:tml><body><h1>400\x20Bad\x20Request</h1></body></html>")%r(DNSStatusReq
SF:uestTCP,B7,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERVER:\x20Linux,\x20
SF:UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20devices/2\.0\r\nCONTENT
SF:-LENGTH:\x2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<html><body><h1>400
SF:\x20Bad\x20Request</h1></body></html>")%r(Help,B7,"HTTP/0\.0\x20400\x20
SF:Bad\x20Request\r\nSERVER:\x20Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x2
SF:0for\x20UPnP\x20devices/2\.0\r\nCONTENT-LENGTH:\x2050\r\nCONTENT-TYPE:\
SF:x20text/html\r\n\r\n<html><body><h1>400\x20Bad\x20Request</h1></body></
SF:html>")%r(SSLSessionReq,B7,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERVE
SF:R:\x20Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20devices
SF:/2\.0\r\nCONTENT-LENGTH:\x2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<ht
SF:ml><body><h1>400\x20Bad\x20Request</h1></body></html>")%r(TerminalServe
SF:rCookie,B7,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERVER:\x20Linux,\x20
SF:UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20devices/2\.0\r\nCONTENT
SF:-LENGTH:\x2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<html><body><h1>400
SF:\x20Bad\x20Request</h1></body></html>");
MAC Address: 68:9E:29:3A:F5:20 (zte)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.00 seconds
```

Figure 4 — nmap -sV 192.168.11.1: service fingerprinting output

3.5 OS detection — nmap -O 192.168.11.1

The OS detection scan identified the router as running Linux kernel 3.2 to 4.14, which is typical for embedded networking hardware like ZTE routers. Nmap classified it as a general purpose device, 1 hop away from our machine.

- Device type: general purpose
- OS: Linux 3.2 - 4.14 (kernel)
- MAC Address: 68:9E:29:3A:F5:20 (ZTE)
- Network Distance: 1 hop

```
C:\Windows\System32>nmap -O 192.168.11.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 01:19 +0000
Nmap scan report for 192.168.11.1
Host is up (0.0032s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE    SERVICE
23/tcp    filtered telnet
53/tcp    open     domain
80/tcp    open     http
443/tcp   open     https
52869/tcp open     unknown
MAC Address: 68:9E:29:3A:F5:20 (zte)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.94 seconds

C:\Windows\System32>
```

Figure 5 — nmap -O 192.168.11.1: OS detected as Linux 3.2-4.14

3.6 Aggressive scan — nmap -A 192.168.11.1

The aggressive scan combines OS detection, service version detection, script scanning and traceroute in one pass. It took 32.64 seconds and confirmed everything found in the previous scans, with extra detail. The traceroute showed the router is exactly 1 hop away (3.11ms RTT), which is expected since it's our gateway.

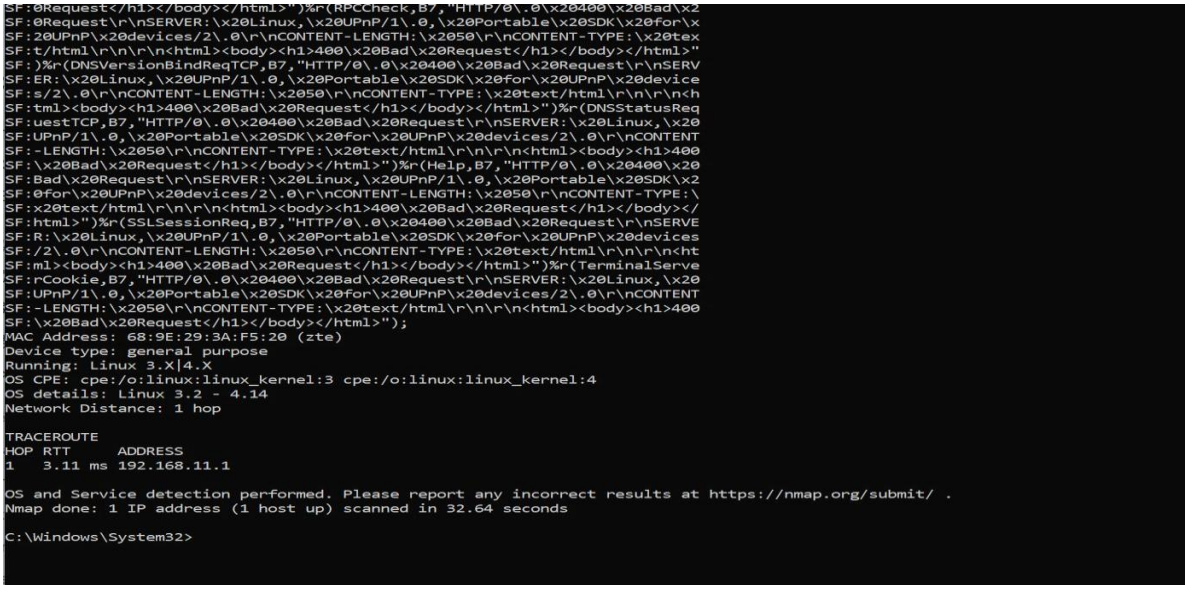
The HTTP server on port 80 immediately redirects to HTTPS. The HTTPS interface returns a full login page with security cookies (SID_HTTPS) and headers indicating it's a web-managed router interface. Port 52869 is confirmed as a UPnP endpoint running on Linux/UPnP 1.0.

```
C:\Windows\System32>nmap -O 192.168.11.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 01:19 +0000
Nmap scan report for 192.168.11.1
Host is up (0.0032s latency).
Not shown: 995 closed tcp ports (reset)
PORT       STATE    SERVICE
23/tcp     filtered telnet
53/tcp     open     domain
80/tcp     open     http
443/tcp    open     https
52869/tcp open      unknown
MAC Address: 68:9E:29:3A:F5:20 (zte)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.94 seconds

C:\Windows\System32>nmap -A 192.168.11.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 01:20 +0000
Nmap scan report for 192.168.11.1
Host is up (0.0031s latency).
Not shown: 995 closed tcp ports (reset)
PORT       STATE    SERVICE    VERSION
23/tcp     filtered telnet
53/tcp     open     domain     (generic dns response: SERVFAIL)
| fingerprint-strings:
|   DNSVersionBindReqTCP:
|     version
|_    bind
80/tcp     open     http
|_http-server-header: <empty>
| fingerprint-strings:
|   GetRequest:
|     HTTP/1.0 302 Moved Temporarily
|     Connection: close
|     Cache-Control: no-cache,no-store
|     Pragma: no-cache
```

Figure 6 — nmap -A 192.168.11.1: aggressive scan, beginning of output

```
|     Connection: close
|     Cache-Control: no-cache,no-store
|     Pragma: no-cache
|     Content-Length: 171593
|     Set-Cookie: SID_HTTPS_=55c9d24dcc7898de28b845ce90e6a2ccb662d9360a419c364a760618803a1174; PATH=/; Secure; H
=
|     Set-Cookie: _TESTCOOKIESUPPORT_HTTPS_=1; PATH=/; Secure; HttpOnly; SameSite=strict
|     Server:
|     Accept-Ranges: bytes
|     X-Content-Type-Options: nosniff
|     X-XSS-Protection: 1; mode=block
|     Content-Security-Policy: frame-ancestors 'self'
|     X-Frame-Options: SAMEORIGIN
|     Content-Type: text/html; charset=utf-8
|     Strict-Transport-Security: max-age=31536000; includeSubDomains; preload
|     <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/transitional.dt
|     <html lang="en" xmlns="http://www.w3.org/1999/xhtml">
|     <head>
|     <meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
|_    <meta http-equiv="X-UA-Compatible" cont
|_http-server-header: <empty>
|_http-title: 400 Bad Request
52869/tcp open      unknown
| fingerprint-strings:
|   DNSStatusRequestTCP, DNSVersionBindReqTCP, Help, RPCCheck, RTSPRequest, SSLSessionReq, TerminalServerCookie:
|     HTTP/0.0 400 Bad Request
|     SERVER: Linux, UPnP/1.0, Portable SDK for UPnP devices/2.0
|     CONTENT-LENGTH: 50
|     CONTENT-TYPE: text/html
|     <html><body><h1>400 Bad Request</h1></body></html>
|   HTTPOptions:
|     HTTP/0.0 501 Not Implemented
|     SERVER: Linux, UPnP/1.0, Portable SDK for UPnP devices/2.0
|     CONTENT-LENGTH: 54
|     CONTENT-TYPE: text/html
|_    <html><body><h1>501 Not Implemented</h1></body></html>
4 services unrecognized despite returning data. If you know the service/version, please submit the following fin
ap.org/cgi-bin/submit.cgi?new-service :
==============NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)==============
SF-Port53-TCP:V=7.98%I=7%D=3/25%Time=69C33873%P=i686-pc-windows-windows%r(
SF:DNSVersionBindReqTCP,20,"\0\x1e\0\x06\x81\x82\0\x01\0\0\0\0\0\x07vers
SF:ion\x04bind\0\0\x10\0\x03");
```

Figure 7 — nmap -A: HTTPS interface details and security headers



```
SF:0Request</h1></body></html>")%r(RPCCheck,B7,"HTTP/0\.0\x20400\x20Bad\x2
SF:0Request\r\nSERVER:\x20Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x20for\x
SF:20UPnP\x20devices/2\.0\r\nCONTENT-LENGTH:\x2050\r\nCONTENT-TYPE:\x20tex
SF:t/html\r\n\r\n<html><body><h1>400\x20Bad\x20Request</h1></body></html>"
SF:)%r(DNSVersionBindReqTCP,B7,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERV
SF:ER:\x20Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20device
SF:s/2\.0\r\nCONTENT-LENGTH:\x2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<h
SF:tml><body><h1>400\x20Bad\x20Request</h1></body></html>")%r(DNSStatusReq
SF:uestTCP,B7,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERVER:\x20Linux,\x20
SF:UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20devices/2\.0\r\nCONTENT
SF:-LENGTH:\x2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<html><body><h1>400
SF:\x20Bad\x20Request</h1></body></html>")%r(Help,B7,"HTTP/0\.0\x20400\x20
SF:Bad\x20Request\r\nSERVER:\x20Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x2
SF:0for\x20UPnP\x20devices/2\.0\r\nCONTENT-LENGTH:\x2050\r\nCONTENT-TYPE:\
SF:x20text/html\r\n\r\n<html><body><h1>400\x20Bad\x20Request</h1></body></
SF:html>")%r(SSLSessionReq,B7,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERVE
SF:R:\x20Linux,\x20UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20devices
SF:/2\.0\r\nCONTENT-LENGTH:\x2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<ht
SF:ml><body><h1>400\x20Bad\x20Request</h1></body></html>")%r(TerminalServe
SF:rCookie,B7,"HTTP/0\.0\x20400\x20Bad\x20Request\r\nSERVER:\x20Linux,\x20
SF:UPnP/1\.0,\x20Portable\x20SDK\x20for\x20UPnP\x20devices/2\.0\r\nCONTENT
SF:-LENGTH:\x2050\r\nCONTENT-TYPE:\x20text/html\r\n\r\n<html><body><h1>400
SF:\x20Bad\x20Request</h1></body></html>");
MAC Address: 68:9E:29:3A:F5:20 (zte)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   3.11 ms 192.168.11.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 32.64 seconds

C:\Windows\System32>
```

Figure 8 — nmap -A: OS confirmation and traceroute result

4. Identified Hosts and Services

Summary of everything discovered during the lab:

IP Address	MAC / Vendor	Known Info
`192.168.11.1`	ZTE	Router/gateway — Linux, ports 53/80/443/52869 open
`192.168.11.100`	Unknown	Active device, not scanned in detail
`192.168.11.101`	Unknown	Active device, not scanned in detail
`192.168.11.102`	Unknown	Active device, not scanned in detail
`192.168.11.103`	Our PC	Windows 11 — Alienware, our scanning machine
`192.168.11.104`	Unknown	Active device, not scanned in detail

5. Analysis Questions

Question 1 — How many hosts were detected?

6 hosts were detected active on the 192.168.11.0/24 network. This includes the router (192.168.11.1), our own machine (192.168.11.103), and 4 other devices — likely phones or other computers connected to the same Wi-Fi.

Question 2 — Which ports are most common?

The most commonly open ports observed on the router were:

- Port 53 (DNS) — present on virtually every router since it handles name resolution

- Port 80 (HTTP) — web management interface, even if it redirects to HTTPS
- Port 443 (HTTPS) — secure web management interface

Port 23 (Telnet) was present but filtered, meaning it's blocked at the firewall level. These are exactly the ports you'd expect on any home/campus router.

Question 3 — Why are open ports a risk?

Every open port is a potential entry point. Here's why it matters:

- Port 80/443 expose the router's admin interface to anyone on the network. If the password is weak or default, someone could take control of the router.
- Port 53 running DNS means someone could potentially query or abuse the DNS resolver.
- Port 52869 (UPnP) is notoriously exploitable — UPnP has a long history of vulnerabilities and is often abused to open ports on routers without authentication.

The general principle: the more ports open, the larger the attack surface. Every open port that isn't strictly necessary should be closed.

Question 4 — What information can attackers gather?

From what we gathered in this lab alone, an attacker could already know:

- The exact number of devices on the network and their IP addresses
- Which device is the router and what brand it is (ZTE) — useful for looking up known CVEs
- The OS running on the router (Linux 3.2-4.14) — narrows down exploit options
- Which services are accessible and their versions — enables targeted attacks
- That the router has an HTTP/HTTPS admin panel — a direct target for credential brute-forcing
- UPnP is running — a common vector for network pivoting and port redirection attacks

6. Security Observations

A few things stood out during this lab:

- The whole network was mapped in under 4 seconds. Just a ping scan and we already know how many devices are connected, their IPs and MAC addresses. That's a lot of information for almost no effort.
- The router runs an admin interface accessible on port 80/443 by anyone on the Wi-Fi. If the default credentials were never changed, an attacker on the network could take over the router completely.
- UPnP on port 52869 is a real concern. UPnP is designed for convenience but it has a well-documented history of being exploited. Most security guides recommend disabling it entirely.
- OS fingerprinting works even without admin access. Just from TCP responses, Nmap identified the kernel version range. This kind of information helps attackers choose the right exploits.
- Ethical reminder from the lab: all of this was done on a network we're authorized to use. Running these scans on any other network without explicit permission is illegal.

7. Conclusion

This lab gave me a very concrete picture of what network reconnaissance actually looks like. Before this I knew theoretically that attackers "scan networks," but seeing it live — 6 hosts discovered in 4 seconds, OS identified, open ports listed — makes the threat much more real.

The most important takeaway for me as a developer is that anything you expose on a network — a port, a service, an admin panel — becomes a potential target. Nmap showed us that gathering this information requires almost no skill or time. The bar for an attacker to do basic recon is extremely low.

Going forward, every application I build that has any network-facing component needs to follow the principle of least exposure: only open what's strictly necessary, keep everything else closed or filtered.